IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 22 September 2025, accepted 12 October 2025, date of publication 16 October 2025, date of current version 23 October 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3622350

RESEARCH ARTICLE

Multi-Step LLM Pipeline for Enhancing TTP Extraction in Cyber Threat Intelligence

HYOUNG ROK KIM[1,2], DONGHYEON LEE[3], INSUP LEE[1], SOOHAN LEE[2], AND SANGJIN LEE[1]

[1]School of Cybersecurity, Korea University, Seoul 02841, Republic of Korea
[2]Ministry of National Defense, Seoul 04383, Republic of Korea
[3]Department of Computer Science and Engineering, Korea University, Seoul 02841, Republic of Korea

Corresponding author: Sangjin Lee (sangjin@korea.ac.kr)

This work was supported by Korea University Grant.

ABSTRACT Tactics, techniques, and procedures (TTPs) are essential for modeling adversary behavior and supporting cyber defense operations. Despite their importance, most cyber threat intelligence (CTI) is provided in unstructured formats, making automated TTP extraction challenging. While manual identification is labor-intensive, current automated approaches suffer from limited accuracy and coverage. To address these challenges, we present a novel multi-step framework based on large language models (LLMs) for extracting MITRE ATT&CK techniques from raw CTI document. Our framework consists of three components: an LLM-based *Extractor* for extracting procedure-level threat actions, an embedding-driven *Technique Candidate Generator* for retrieving semantically relevant technique candidates, and a *Validator* that ranks candidate techniques by likelihood using LLM inference to refine final predictions and reduce false positives. Experimental results on the benchmark dataset demonstrate that our approach significantly outperforms existing baselines, achieving an F1-score of 82.28%, thereby validating its effectiveness. Additionally, the modularity of our framework allows seamless integration of future LLMs, suggesting continual performance gains as foundation models evolve.

INDEX TERMS Cyber threat intelligence, MITRE ATT&CK, large language model, information retrieval.

I. INTRODUCTION

Advanced persistent threats (APTs) are sophisticated cyber-attacks marked by prolonged, covert infiltration into targeted systems, with the goal of stealing sensitive information or disrupting critical operations [1], [2]. While traditional defense mechanisms rely heavily on static Indicators of Compromise (IoCs) such as malicious IPs and file hashes [3], these indicators are easily obfuscated or modified, limiting their long-term effectiveness [4]. In contrast, analyzing the behavioral characteristics of attackers by identifying their utilized tactics, techniques, and procedures (TTPs) can provide deeper insights into their underlying attack methodologies [5]. TTPs represent consistent behavioral patterns exhibited by attackers over time, and once identified, they remain valuable for extended periods. Recognizing this advantage, cybersecurity professionals are increasingly adopting behavior-based modeling approaches centered around TTPs, notably exemplified by standard frameworks such as MITRE ATT&CK [6].

Despite the growing importance of TTPs, a major challenge lies in extracting them from cyber threat intelligence (CTI) sources [8]. APT reports released by cybersecurity vendors are predominantly available in unstructured formats, such as PDFs or HTML documents. Although a substantial number of such reports are published, only a small fraction are transformed into structured TTP representations [9]. This gap primarily stems from the requirement of specialized domain knowledge and the significant manual effort involved in identifying and annotating relevant TTPs. As a consequence, the conversion process is both time-consuming and difficult to scale, hindering the ability of analysts to maintain comprehensive and up-to-date TTP knowledge bases.

To address this issue, various research efforts have been conducted on automatically extracting TTPs from unstructured CTI using a range of methods, including traditional text

The associate editor coordinating the review of this manuscript and approving it for publication was Thomas Canhao Xu.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

mining techniques, natural language embedding techniques, and deep learning-based models [11], [12], [13], [14], [15], [16]. Early studies focused on classifying malicious behavior using traditional feature extraction methods [11] such as named entity recognition (NER), but were limited to explicitly defined entities, missing broader semantic contexts. Traditional text embedding methods, such as latent semantic indexing (LSI), were later introduced but struggled to capture the complexity of attack behaviors in documents [13]. With advancements in deep learning, models like BERT enabled context-aware embeddings. However, their effectiveness was constrained by limited TTP-labeled data, leading to inaccurate extractions and a reliance on human validation [14], [18]. Recently, large language models (LLMs) have shown significant promise in understanding attacker behaviors within CTI documents due to their advanced contextual and situational comprehension [37]. Nonetheless, LLMs sometimes produce inaccurate information, a phenomenon known as hallucination [26].

To address the limitations of relying solely on LLMs, we propose a hybrid framework that combines LLMs with sentence embeddings and semantic similarity search. This approach mitigates hallucination-induced inaccuracies and improves extraction precision by leveraging the complementary strengths of generative AI and structured knowledge bases. Specifically, by providing LLMs with candidate TTPs retrieved from the knowledge base, our method achieves more reliable technique identification. Furthermore, our framework is designed for adaptability, enabling continuous updates to TTP information based on the latest STIX data maintained by MITRE [27]. This ensures that security analysts can access current and accurate threat intelligence, effectively addressing the dynamic nature of cybersecurity threats.

The key contributions of our research include:

- We design a multi-stage LLM pipeline that fully leverages the contextual understanding capabilities of LLMs and integrates sentence embedding-based similarity search techniques to overcome the limitations of LLM-only approaches.
- We conduct comprehensive performance evaluations to demonstrate that our framework surpasses existing state-of-the-art methodologies and baseline LLM approaches in extracting MITRE ATT&CK techniques from unstructured CTI documents.
- We experimentally show that the effectiveness of our framework stems from two key factors: the superior semantic representation provided by LLM-based embeddings compared to BERT-based embeddings, and the precision gains achieved by validating candidate techniques using LLM, which effectively reduces false positives.

The remainder of this paper is organized as follows: Section II introduces the background knowledge essential for understanding our work. Section III reviews related studies on TTP extraction from unstructured text. Section IV describes our proposed framework, detailing methods for threat action extraction, the construction of a threat action embedding space, and the replacement of human validation with LLMs. Section V compares existing approaches with the performance of our framework, while Section VI evaluates the contribution of each framework component. Finally, Section VII concludes the paper with a discussion on future research directions.

II. BACKGROUND

This section provides the foundational knowledge necessary to understand our research. We focus on the MITRE ATT&CK framework for standardizing TTPs and introduce LLMs, their distinctive strengths, and their potential applications in cybersecurity threat intelligence.

A. MITRE ATT&CK FRAMEWORK

MITRE ATT&CK is a knowledge-based framework that systematically documents the TTPs of real-world cyber adversaries. This framework establishes a practical standard for understanding attacker behavior patterns and developing effective defense strategies. Unlike earlier, more generalized models such as the Lockheed Martin Cyber Kill Chain [7], ATT&CK offers a significantly finer-grained and comprehensive taxonomy of adversary actions. ATT&CK organizes adversarial actions into a hierarchical structure, categorized into tactics, techniques, and procedures. Tactics represent the high-level objectives of an attacker (e.g., initial access, privilege escalation), while techniques describe the methods used to achieve these goals, each identified by a unique ID (e.g., T1566). Sub-techniques are specific variants of a parent technique. For instance, the phishing technique (T1566) is further divided into spear phishing attachment (T1566.001), spear phishing link (T1566.002), and so forth. Procedures refer to concrete, real-world implementations of techniques, typically observed in specific threat actor campaigns or malware behaviors, and provide detailed insight into how techniques are executed in practice.

ATT&CK's global adoption serves as a critical standard for Security Operations Center (SOC) activities and incident response, reinforcing its practical applicability in cybersecurity operations. Fig. 1 illustrates a comprehensive workflow that begins with APT attack occurrences, progresses through CTI documentation by cybersecurity companies, and is transformed into MITRE TTPs through extraction systems such as the one proposed in this paper. These extracted TTPs subsequently enable various cybersecurity applications. For threat intelligence sharing, platforms such as MISP [28] and OpenCTI [29] incorporate MITRE TTPs to structure their databases, enabling efficient information exchange between organizations. Additionally, automated attack emulation frameworks such as CALDERA [17] utilize MITRE TTPs to simulate realistic threats and evaluate an organization's response capabilities.

The utility of MITRE ATT&CK is further enhanced when integrated with Structured Threat Information eXpression

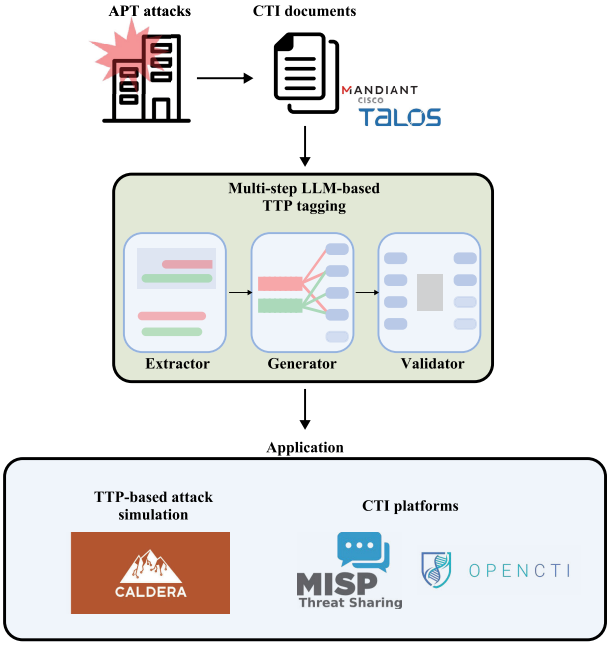


FIGURE 1. System overview. Following APT attacks, security vendors generate CTI documents to analyze adversary behaviors, which serve as sources for extracting TTPs. Our multi-step LLM-based pipeline can effectively automate the TTP extraction process, enabling downstream applications such as TTP-based attack simulation and CTI platforms.

(STIX) format [10], a standardized language for sharing cybersecurity intelligence. STIX represents CTI as serializable JSON, which can include ATT&CK TTPs within its structured data. This facilitates threat information sharing between organizations and enhances interoperability among analysis platforms. For example, through STIX, ATT&CK techniques can be connected with other threat intelligence data to enrich the context of security events. MITRE ATT&CK is regularly updated and provides matrices for various platforms including Enterprise, Mobile, and Industrial Control Systems, ensuring its continued relevance and accuracy in a rapidly evolving threat landscape. Our research aims to enhance the precision of threat analysis by focusing on TTP extraction with a technique-specific approach based on the MITRE ATT&CK framework.

B. LARGE LANGUAGE MODEL IN CYBER THREAT INTELLIGENCE

Recent advances in natural language processing (NLP) have rapidly progressed with the emergence of pre-trained language models of increasing scale and capability. These models demonstrate exceptional performance in effectively learning language representations from vast training data, generating and understanding natural text [36].

The development of large-scale language models has evolved significantly, beginning with encoder-based models such as BERT [38]. These models utilize the encoder structure of the transformer architecture to effectively convert text meanings into vector representations, showing specialized performance in natural language understanding tasks. In the CTI field, models have emerged that are trained domain-specifically on text collected from various CTI sources, including APT reports, and these models have demonstrated promising performance on various NLP tasks in cybersecurity [23], [34].

Building upon these foundations, decoder-based LLMs like the GPT series have further advanced the field by utilizing the decoder structure of the transformer architecture to enable natural text generation based on previous context [35]. These models, typically containing billions of parameters and significantly larger than their encoder-based counterparts, possess remarkable few-shot learning capabilities that allow quick adaptation to new tasks and domains with minimal example data, as well as zero-shot learning abilities that can solve new problems without additional training data, further expanding their potential applications. Decoder-based LLMs have recently gained significant research interest in the CTI field due to their natural language understanding, generation, and summarization capabilities, which are particularly effective for analyzing vast text-based threat data [37]. Recent efforts have focused on developing domain-specific LLMs for cybersecurity to further enhance threat intelligence analysis and automation [33]. These models are increasingly being evaluated through specialized benchmarks that assess their ability to perform practical cybersecurity tasks such as vulnerability assessment, threat attribution, and technique extraction [21].

Motivated by this background, this research aims to enhance the accuracy of MITRE ATT&CK technique extraction by leveraging decoder-based LLMs. Furthermore, we comparatively evaluate our proposed method with existing encoder-based pre-trained models, demonstrating the effectiveness and advantages of our approach in addressing this critical cybersecurity task.

III. RELATED WORK

The extraction of TTPs from unstructured CTI documents has been a critical research area in cybersecurity. This section reviews prior studies on TTP extraction, categorizing them into two primary approaches: ontology-based attack pattern extraction and classification-based technique identification. We discuss the key contributions and limitations of each approach to contextualize our proposed framework.

A. ONTOLOGY-BASED ATTACK PATTERN EXTRACTION

Ontology-based methods leverage domain-specific knowledge to extract structured attack patterns from CTI documents. Husari et al. [22] introduced TTPDrill, a pioneering approach that employs NLP techniques, specifically Part-of-Speech (POS) parsing, alongside a threat ontology to identify attack patterns from unstructured CTI sources. Similarly, Rahman and Williams [13] evaluated multiple extraction techniques, including LSI, subject-verb-object (SVO) tuple extraction, and BM25 ranking functions. However, these methods often lack specificity in their classification

granularity, which limits their ability to cover the full spectrum of attack techniques in the MITRE ATT&CK framework (e.g., Rahman and Williams [13] only covered 64 classes out of the hundreds defined in the framework).

More recent efforts include AttacKG [20], which constructs technique knowledge graphs from CTI reports by extracting attack-relevant entities and dependencies using NER. The system identifies techniques using a graph alignment algorithm that calculates node similarities with technique templates. While effective for visualizing attack patterns as knowledge graphs, AttacKG's pre-initialized templates lack sufficient information to identify specific techniques, particularly when confronted with novel or significantly modified attack variants. LADDER [25] improves upon AttackG by using sequence tagging techniques (BRAT) to extract entities and relationships, enhancing attack strategy categorization. Nevertheless, its accuracy remains lower than models that process entire sentences, as it primarily uses extracted patterns for similarity measurement [19]. Li et al. [14] combined BERT with BiLSTM and conditional random field (CRF) to extract threat actions, improving efficiency over manual processes. However, this approach struggles with complex or multifaceted threat actions within single sentences.

Recent studies have also explored LLMs for CTI tasks [21], [33]. These efforts mainly focused on measuring LLMs' cybersecurity capabilities and highlight their potential in handling complex threat intelligence, but face challenges in practical implementation due to reliability issues, such as hallucination [33]. Overall, ontology-based approaches demonstrate substantial performance at identifying threat entities but often exhibit lower accuracy in precise TTP mapping due to limited contextual understanding.

B. CLASSIFICATION-BASED TECHNIQUE IDENTIFICATION

Classification-based methods utilize supervised learning to map sentences from CTI documents to specific TTPs. rcATT [12] employed Term Frequency-Inverse Document Frequency (TF-IDF) to vectorize words in CTI documents, and then evaluated several traditional machine learning algorithms, including Naive Bayes, SVM, Random Forest, and Logistic Regression. While this approach achieved moderate accuracy, its performance was constrained by the limitations of the underlying embedding representation.

To address these limitations, MITRE's TRAM [18] incorporated SciBERT [24], a BERT model pre-trained on a large corpus of scientific publications, to enhance the contextual representation of CTI documents. As a result, TRAM achieved improved classification performance on MITRE ATT&CK techniques. However, its applicability remains confined to a subset of frequently occurring techniques. Orbinato et al. [15] fine-tuned SecureBERT [23], a BERT model trained on cybersecurity-specific textual data. By utilizing a model trained on domain-specific data, they successfully expanded the classification coverage to 188 MITRE ATT&CK techniques, outperforming prior models in terms of scope. Nonetheless, it faces challenges with varying optimal threshold values across different CTI documents. Li et al. [16] enhanced TTP classification using DistilBERT with a post-processing step that exploits tactic-technique relationships, achieving higher accuracy but lower generalization across diverse datasets. TTPXHunter [19], the current state-of-the-art for document-level technique extraction, improves performance by augmenting datasets with Masked Language Modeling (MLM) targeting the full ATT&CK framework. Despite its advancements, it relies on arbitrary confidence thresholds and softmax, limiting generalization and requiring human intervention.

Classification-based methods generally outperform ontology based approaches by processing entire sentences, enhancing their ability to distinguish between techniques by utilizing a broader range of information. However, their generalization performance often decreases significantly across different datasets, as evidenced by a reported accuracy drop from 86.37% to 38.05% in cross-validation experiments between ATT&CK and TRAM datasets [16].

In summary, ontology-based methods struggle with comprehensive semantic understanding, leading to lower technique mapping accuracy, while classification-based approaches offer improved performance but face challenges in generalization and require manual thresholding. LLM-based approaches, despite their recent promise in contextual understanding, are hampered by reliability issues. These collective limitations underscore the need for a hybrid approach that combines the contextual understanding of LLMs with structured knowledge bases, which our proposed framework aims to address.

IV. PROPOSED METHODOLOGY

In this section, we propose a multi-step LLM pipeline designed to effectively identify MITRE ATT&CK techniques from CTI documents.

To address the limitations of ontology-based and classification-based approaches, we introduce a hybrid framework that combines LLMs' advanced contextual understanding with a sentence embedding-based similarity search. The proposed framework consists of three components that operate sequentially: an *Extractor*, a *Technique Candidate Generator*, and a *Validator*. A schematic overview of this architecture is presented in Fig. 2, while Table 1 details the specific role, input, and output of each component. The *Extractor* first extracts complete sentences describing attack behaviors from CTI documents. Next, the *Technique Candidate Generator* retrieves candidate techniques by employing a semantic similarity method built on our MITRE ATT&CK based embedding space. Finally, the *Validator* component uses an additional LLM to refine these candidate techniques, ensuring high-accuracy identification. In the following subsections, we provide an in-depth description of the design, implementation, and optimization of each component.

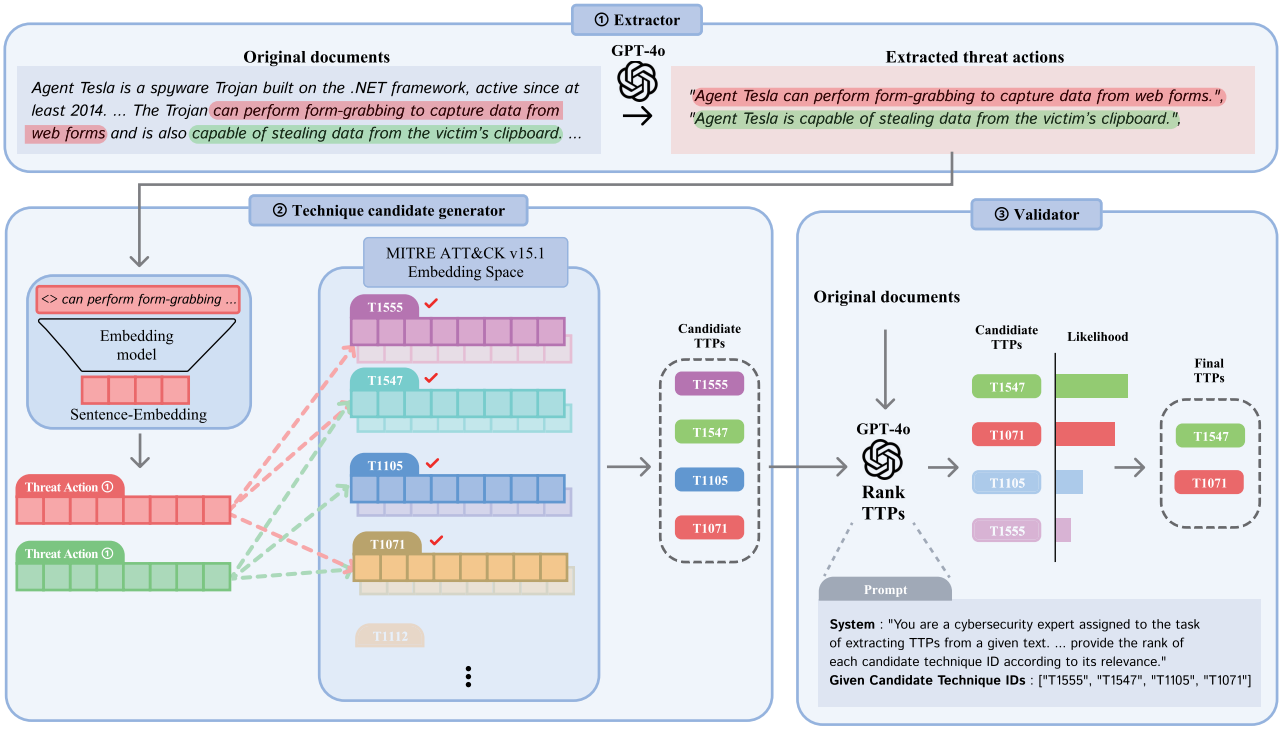


FIGURE 2. An overview of multi-step LLM pipeline for TTP extraction consisting of three components: (i) the *Extractor* identifies threat actions from input text; (ii) the *Technique Candidate Generator* maps the extracted sentences into a MITRE ATT&CK technique embedding space; and (iii) the *Validator* ranks the candidate techniques and identifies the most appropriate technique IDs (e.g., T1547, T1071). The example illustrates how the description of "Agent Tesla," a spyware Trojan, is processed through the pipeline to identify precise techniques IDs.

TABLE 1. Component specifications of multi-step LLM pipeline.

Component	Input	Output	Function
Extractor	Raw CTI Document	Threat Actions	Extracts procedure-level threat behaviors.
Candidate Generator	Threat Actions (from *Extractor*), ATT&CK v15.1 Knowledge Base	Candidate TTPs	Generates a set of relevant TTP candidates by comparing the semantic similarity between threat actions and procedures in the ATT&CK v15.1 knowledge base.
Validator	Candidate TTPs (from *Generator*), CTI Document	Finalized TTPs	Evaluate and rank candidate TTPs based on their likelihood of being present in the CTI document.

A. EXTRACTOR

The initial component of our framework, the *Extractor*, is designed to identify and extract MITRE ATT&CK procedure-level threat actions from unstructured CTI text. The key innovation of our *Extractor*, which leverages LLM, is its ability to process complex text descriptions and generate well-formed sentences that each correspond to a single MITRE ATT&CK technique. Our approach utilizes LLM's generative capabilities to produce threat actions in our desired structured format, enabling more precise mapping to MITRE ATT&CK techniques.

As illustrated in Fig. 3, when given a paragraph describing specific malware, our *Extractor* precisely isolates individual sentences that map to distinct techniques defined in the MITRE ATT&CK framework. As observed in [14], many threat descriptions in CTI documents contain sentences that inherently reference multiple techniques. To address this issue, we instruct the *Extractor* to decompose such composite sentences into multiple, technique-specific ones, where each resulting sentence corresponds to a single technique. This decomposition process enhances the accuracy of subsequent embedding-based candidate technique generation steps.

Another significant advantage of our *Extractor* is that it minimizes information loss. This approach represents a clear departure from previous methods that relied primarily on keyword extraction, often producing short, fragmented phrases (e.g., "creates file" or "Trojan creates registry entries") [11]. Such fragmented extractions frequently lead to ambiguity due to the loss of contextual information. For instance, a simple phrase like "creates file" could

Prompt : "You are a helpful AI assistant specialized in cybersecurity. Extract sentences from the given text that are valuable for TTP extraction in relation to malware behavior, and respond with a numbered Python list format. **If a single sentence includes multiple techniques, please split them so that each technique is on its own line, even if they come from the same sentence.**"

Given text : 4H RAT is a piece of malware linked to Putter Panda, with evidence of its use dating back to at least 2007. It relies on HTTP for command and control (C2) communication and has the ability to create a remote shell...

Extracted threat actions by LLM:
1. 4H RAT relies on HTTP for command and control (C2) communication.
2. 4H RAT has the ability to create a remote shell.
3. ...

FIGURE 3. An example of threat action extraction. The *Extractor* takes as input a malware description of "4H RAT" and outputs a list of sentences, which are formatted to facilitate efficient processing by subsequent components of the framework.

...,"description": "[3PARA RAT](https://attack.mitre.org/software/S0066) command and control commands are encrypted within the HTTP C2 channel using the DES algorithm in CBC mode with a key derived from the MD5 hash of the string HYF54&%9&jkMCXuiS. [3PARA RAT](https://attack.mitre.org/software/S0066) will use an 8-byte XOR key derived from the string HYF54&%9&jkMCXuiS if the DES decoding fails (Citation: CrowdStrike Putter Panda)", "relationship_type" : "uses", "source_ref": "malware– 7bec698a-7e20-4fd3-bb6a-12787770fb1a", "target_ref": "attack-pattern–24bfaeba-cb0d-4525-b3dc- 507c77ecec41", "x_mitre_version": "1.0", "x_mitre_modified_by_ref": "identity–c78cb6e5-0c4b-4611-8297-d1b8b55e40b5", ...

FIGURE 4. An example of STIX-formatted data from the MITRE ATT&CK framework. To construct the knowledge base, threat procedures are extracted from the "description" field, while the corresponding technique ID is obtained from the "target_ref" field.

ambiguously correspond to multiple techniques such as "T1150 (Ingress Tool Transfer)", "T1074 (Data Staged)", or "T1543 (Create or Modify System Process)", making precise identification more difficult. However, our *Extractor* effectively utilizes LLMs to generate complete, context-rich sentences representing threat actions with minimal information loss. Beyond preserving information integrity, while keyword approaches are limited to explicit matches, our method can also capture implicit meanings through LLM's semantic understanding capabilities.

Section VI-A provides a detailed evaluation of our *Extractor*'s performance and effectiveness.

B. TECHNIQUE CANDIDATE GENERATOR

The second component of our pipeline, the *Technique Candidate Generator*, identifies suitable candidate techniques corresponding to input sentences by leveraging sentence embedding similarity with a pre-constructed embedding space. This step bridges the gap between textual threat actions extracted from CTI documents and formalized MITRE ATT&CK techniques. The process of generating candidate techniques involves establishing a robust embedding space, followed by an embedding-based semantic similarity search. We describe these two stages in detail below.

1) CONSTRUCTION OF EMBEDDING SPACE

To effectively match extracted threat actions with relevant MITRE ATT&CK techniques based on their semantic meaning, we constructed a comprehensive embedding knowledge base. This knowledge base explicitly maps procedures to their corresponding unique technique ID. For this mapping, we extracted procedures and technique IDs from version 15.1 of the MITRE ATT&CK knowledge base, which MITRE publicly releases in STIX 2.0 format [27]. Specifically, as shown in Fig. 4, we obtained procedures from the *"description"* field and extracted the attack pattern identifiers from the *"target_ref"* field. Using the MITRE STIX dataset, we then converted these attack pattern identifiers to their corresponding MITRE ATT&CK technique IDs.

For implementation, we parsed the STIX-formatted JSON files using the python-stix2 library [43]. We filtered STIX entries to retain only those labeled as *"relationship"* and excluded any whose source reference was labeled as *"course-of-action"*, thereby focusing solely on procedures representing attacker behaviors. This systematic extraction process ultimately resulted in a total of 13,371 definitive procedure-technique pairs, covering 529 distinct techniques, including sub-techniques.

Semantic similarity between procedures can be distorted when the comparison is overly influenced by named entities, such as specific malware families or threat actor groups. These entities often appear across different techniques and procedures, potentially leading to misleading matches based on shared terminology rather than actual behavioral similarity. To mitigate this issue, we applied a systematic entity-masking process to the procedures in our knowledge base. Specifically, we replaced named entities in the context of CTI, such as malware names (e.g., *Emotet*, *TrickBot*) and threat groups (e.g., *APT28*), with generic placeholders. This preprocessing step ensures that semantic embeddings focus on the tactical and technical aspects of attacker behavior rather than surface-level identifiers. The masked procedures were then embedded using OpenAI's *text-embedding-ada-002* [44], generating 1536-dimensional vector representations. These embeddings were indexed using FAISS [45] to support efficient similarity searches. Section VI-B2 presents a detailed comparison with an encoder-based embedding approach, using sentence-BERT (SBERT) [46] as a representative model.

2) TECHNIQUE CANDIDATE RETRIEVAL

Following the construction of the embedding space, we use similarity-based retrieval to identify candidate techniques for

TABLE 2. Average hit@k performance across different k values.

k	1	2	3	4	5	6	7	8
Hit@k	0.69	0.79	0.83	0.85	0.86	0.88	0.89	0.89

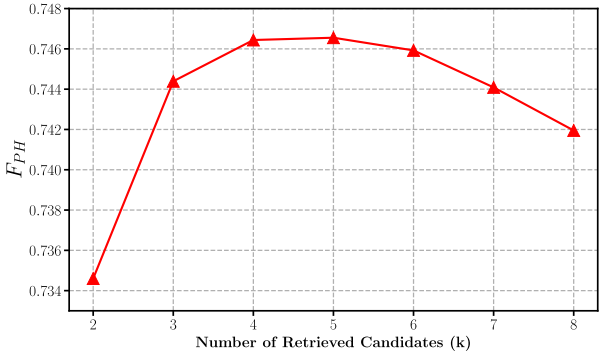


FIGURE 5. F_{PH} scores for different numbers of retrieved candidates (k). The plot demonstrates the trade-off between Precision@k and Hit@k as reflected in the F_{PH} score. The optimal performance is achieved at $k = 5$.

each extracted threat action. Each threat action extracted by the *Extractor* is embedded using *text-embedding-ada-002*, and these embeddings are used to query the indexed embedding space. The retrieval process uses L2 distance to identify the top-k semantically similar procedures from the knowledge base. Each retrieved procedure maps directly to its corresponding technique ID, resulting in a set of candidate techniques for validation.

This approach is motivated by the observation that procedures belonging to the same technique tend to produce similar sentence embeddings. To verify this, we conducted an analysis using the constructed embedding space. We computed the semantic similarity between all procedures in version 15.1 of the MITRE ATT&CK knowledge base using L2 distance between sentence embeddings. Table 2 presents the average hit@k results, where hit@k indicates whether the top-k most similar procedures include at least one procedure sharing the same technique ID as the query. These results provide empirical support for our sentence embedding-based approach. The hit@1 score of 0.69 means that nearly 7 out of 10 cases, the most similar procedure belongs to the same technique, which is considerably high given the 192 distinct techniques in the knowledge base.

To determine the optimal number of similar procedures (k) to retrieve for effective technique selection, we analyzed various top-k settings ranging from 2 to 8. We evaluated two key metrics: precision@k, which represents the proportion of retrieved candidate procedures that share the same technique as the actual procedure, and hit@k, which has the same definition as mentioned above. Here, k denotes the number of top-ranked procedures retrieved based on semantic similarity. To balance these metrics, we calculated a harmonic mean of precision@k and hit@k, which we define as follows to ensure our candidate technique set includes the actual techniques while minimizing the number of unnecessary candidate techniques:

$$F_{PH} = \frac{2 \cdot \text{Precision@}k \cdot \text{Hit@}k}{\text{Precision@}k + \text{Hit@}k} \tag{1}$$

As depicted in Fig. 5, our analysis shows that the F_{PH} score reaches its maximum value at k=5, demonstrating an effective balance between precision@k and hit@k for our technique retrieval process. Therefore, we select the top-5 most similar procedures from our knowledge base to form the candidate technique set, which is then passed to the *Validator* component for final technique selection.

> Prompt : "You are a cybersecurity expert assigned to extract TTPs from a given paragraph. A list of candidate technique IDs that may exist in the paragraph will be provided. Your task is: identify which candidate TTPs are actually present in the paragraph and rank each candidate TTP according to its likelihood of being present in the document. Output the ranked list of candidate TTPs, and ensure that the final line contains only the IDs of the main techniques, separated by commas."
>
> Given Candidate Technique IDs : ["T1555", "T1547", "T1185", "T1071", "T1119", "T1115", "T1552", "T1140", "T1087", "T1573", "T1082", "T1560", "T1027", "T1543"]
>
> Validated Technique IDs by LLM: ["T1547", "T1071", "T1573", "T1115", "T1555", "T1185", "T1140"]

FIGURE 6. An illustration of the prompt used in the *Validator*, presenting the candidate technique IDs provided to the LLM and the corresponding technique IDs selected as output.

C. VALIDATOR

As the final component, *Validator* selects the final technique ID set by verifying the candidate set from the *Technique Candidate Generator*. Given that reliability is critical for practical cybersecurity applications, it is designed to improve the precision of TTPs extracted by our framework, employing LLM in the process. In systems such as MITRE TRAM [18], this validation step is performed by human analysts through manual review of candidate TTPs. This process is time-consuming, and in practice it is not feasible to clearly distinguish among more than 600 ATT&CK TTPs. To address this limitation, we leverage the extensive pre-training of LLMs on large-scale corpora.

The *Validator* uses contextual understanding of the given document to evaluate candidate TTPs while constraining the output to a predefined candidate set, effectively mitigating hallucination risks. The LLM is instructed to rank the techniques by likelihood before final selection. During this process, techniques with a low likelihood of occurrence are automatically excluded from the final technique ID set. This process is illustrated in Fig. 6, which shows the prompt utilized for *Validator*. The prompt provides the candidate technique IDs along with the original text, enabling

the LLM to make informed decisions based on contextual understanding. The LLM outputs a list of technique IDs ranked by likelihood, which are then parsed to extract the final TTP label set.

Our experiments demonstrate that this approach significantly improves precision from 68.77% to 86.14% compared to using candidate techniques directly without validation. This ranking-based filtering process effectively removes false positives that commonly occur in embedding-based retrieval systems, improving precision significantly while minimizing the impact on recall (see Section VI-C for details).

An additional advantage of our design is its modularity: both the *Extractor* and *Validator* components rely on LLMs independently, allowing each to be seamlessly replaced with newer models as they become available. This ensures that the framework can continuously benefit from advances in LLMs without requiring structural modifications.

V. EXPERIMENTS AND RESULTS

This section evaluates the performance of our framework against state-of-the-art methods using a benchmark dataset for technique extraction. In Section V-A, we describe the chosen evaluation dataset and explain the criteria for its selection. Section V-B outlines the experimental setup, including the replication of baseline methods for a fair comparison. Finally, in Section V-C, we present and discuss the results, highlighting the comparative strengths of our proposed framework.

We assess model performance by measuring their ability to accurately identify sets of technique IDs from CTI documents, formulated as a multi-label classification task. Following established practice in the TTP extraction literature, we adopt the micro F1-score [31] as our primary evaluation metric. The micro F1-score represents a global performance measure that aggregates true positives, false positives, and false negatives across all predictions. This provides a robust metric to assess overall model effectiveness, which is particularly important when dealing with imbalanced data. To facilitate a more comprehensive evaluation of model performance, we additionally report both precision and recall scores individually. These complementary metrics enable deeper insights into each model's strengths and weaknesses, with recall assessing the capability to identify true technique IDs and precision focusing on minimizing false positives. Such analysis helps to reveal key distinctions between existing approaches and LLM-based frameworks in practical deployments.

A. BENCHMARK DATASET

To evaluate the TTP extraction capability of our framework, we utilized the CTI-Bench benchmark dataset [21], which is specifically designed to assess the performance of LLMs in the domain of CTI. Among the various tasks provided by CTI-Bench, we employed the CTI-ATE (Attack Technique Extraction) task. The CTI-ATE dataset comprises pairs of paragraphs and corresponding technique IDs. Each paragraph describes malware characteristics, including specific behavioral procedures and additional contextual information, such as the background and circumstances under which the malware was employed. These contextual sentences do not directly describe threat behaviors, enabling the evaluation of our framework's ability to distinguish relevant attack patterns from surrounding background information. The associated technique IDs represent only the main techniques, explicitly excluding sub-techniques. For example, the sub-technique "T1098.004 (Account Manipulation: SSH Authorized Keys)" is mapped solely to its main technique "T1098 (Account Manipulation)." Following prior studies including [25], our evaluation focused exclusively on techniques associated with the Enterprise platform, enabling direct comparison with existing work.

We selected this dataset for several practical reasons. In many CTI reports, authors tend to label only the technique IDs they personally deem important, or they rely on out-of-date ATT&CK references, making these reports unsuitable for use as performance evaluation datasets. By contrast, this dataset explicitly includes all technique IDs present in the texts, as it was constructed by synthesizing TTP-labeled sentences, ensuring that all TTPs in the text are properly labeled and consistently annotated according to MITRE ATT&CK version 15.1.

B. BASELINE METHODS AND EXPERIMENTAL SETUP

To comprehensively evaluate our framework, we selected five baseline methods representing three major approaches to TTP extraction. For classification-based methods, we used TTPXHunter [19] and a self-finetuned SecureBERT classifier [15]. For similarity-based approaches, we chose AttacKG [20] and LADDER [25]. Finally, we included a Single LLM baseline to represent modern generative approaches. For both the Single LLM baseline and the LLM component of our multi-step framework, we employed ChatGPT-4o [30] as the underlying model. This selection was based on benchmark results reported in [21], which show that ChatGPT-4o outperforms ChatGPT-4 [42], Google Gemini 1.5 [39], and Meta LLaMA models, including LLaMA 3-8B and LLaMA 3-70B [40], [41] on the technique extraction task. Our experimental results replicated this performance trend, reinforcing the validity of this choice. Experiments were performed on an Apple M1 Pro machine (8-core processor, 16 GB RAM) with Metal Performance Shaders (MPS) acceleration. We describe the experimental setup for each baseline as follows.

1) CLASSIFICATION-BASED APPROACH

For these baselines, we selected TTPXHunter [19], which outperforms existing classification methods including encoder-based models and traditional machine learning classifiers [12], [15], [18]. Built on SecureBERT, TTPXHunter achieves state-of-the-art results in technique extraction. For our experiments, we adopted the publicly available

TABLE 3. Hyperparameter settings for the SecureBERT-based classifier.

Hyperparameter	Value
Optimizer	Adam
Loss Function	CrossEntropyLoss
Activation Function	ReLU
Batch Size	16
Epoch	10
Learning rate	1e-5
Dropout Rate	0.3
Classification Threshold	0.4

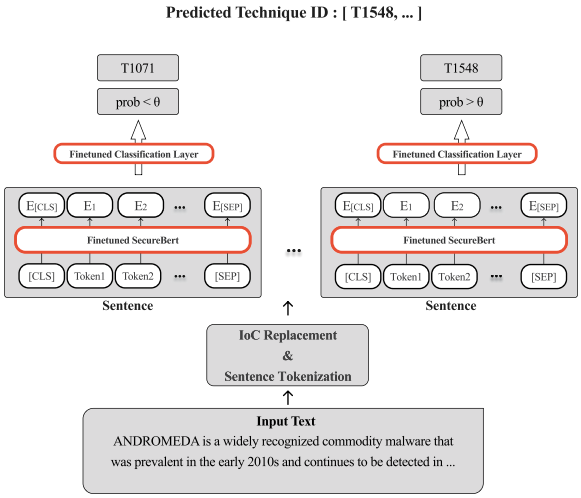


FIGURE 7. Technique extraction method using the SecureBERT-based classifier. First, the text is preprocessed through IoC replacement, and then it is fed into the classifier. If the probability output exceeds a specific threshold, the corresponding technique is selected.

TTPXHunter model from Hugging Face and performed inference using default hyperparameters.

However, due to insufficient information about TTPX-Hunter's training datasets, we added a custom encoder-based classifier to our baselines. This classifier was fine-tuned exclusively on the same knowledge base used in our framework, enabling a more equitable comparison. The model architecture is based on SecureBERT, with additional linear layers for classification. These layers project contextual embeddings into predictions across 192 main techniques included in the benchmark dataset. The hyperparameters used for fine-tuning are summarized in Table 3. The fine-tuning was performed in a Google Colab environment with an A100 GPU. The technique prediction process for the benchmark dataset follows the procedure illustrated in Fig. 7. The input text is preprocessed to normalize IoC expressions and tokenized at the sentence level. Each sentence is then classified using the SecureBERT-based classifier model, and predicted techniques are aggregated to produce the final technique ID set.

2) SIMILARITY-BASED APPROACH

Although similarity-based retrieval methods generally exhibit lower classification performance than encoder-based approaches [19], we included AttacKG and LADDER as baselines due to their conceptual similarity to the retrieval strategy adopted in our framework. AttacKG identifies techniques by comparing extracted attack patterns with predefined, graph-structured templates that represent technique-level knowledge. LADDER performs similarity matching between extracted attack patterns and the MITRE ATT&CK technique descriptions. Notably, we updated AttacKG's technique templates using the latest version of the MITRE ATT&CK procedures for experiments, as the original templates were based on an outdated knowledge base.

TABLE 4. Performance comparison in single-prompt technique extraction settings.

Prompt	Precision	Recall	F1
Vanilla	54.05	66.00	66.69
Tech-Guided	47.69	59.81	59.63
One-Shot	47.05	59.36	60.85
Multi-Label	57.70	70.42	70.56

3) SINGLE LLM BASELINE

We employed a single LLM-based approach as a final baseline. For a fair comparison with our framework, the Single LLM baseline was implemented with the same LLM employed in our framework. To determine the optimal prompt configuration for single LLM-based technique extraction, we conducted a comparative analysis of four different prompt settings.

- **Vanilla Prompt**: The LLM extracts techniques without additional context.
- **Technique-Guided Prompt**: The prompt includes all technique IDs and names from MITRE ATT&CK.
- **One-Shot Prompt**: The model receives a single example of text paired with technique IDs prior to extraction.
- **Sentence-Level Multi-Label Prompt**: The prompt instructs the model to extract all techniques from a sentence containing multiple techniques.

Table 4 shows the precision, recall, and F1-scores for the four prompt settings evaluated on the benchmark dataset. The Vanilla Prompt outperforms both the Technique-Guided and One-Shot Prompts, suggesting that additional information may occasionally reduce extraction performance. This aligns with findings that simpler contexts can be more effective for certain tasks [32]. However, the Vanilla Prompt typically extracts only one technique when multiple techniques occur in a single sentence. To address this limitation, we introduced the Sentence-Level Multi-Label Prompt, which yielded notable performance improvements. Accordingly, we adopted the Sentence-Level Multi-Label Prompt as the standard configuration for the Single LLM Baseline in subsequent evaluations.

C. COMPARISON RESULTS

We conducted a comprehensive evaluation of our proposed technique extraction framework using the benchmark dataset.

TABLE 5. Comparative experiment results. This table presents the extraction results for MITRE ATT&CK technique IDs.

Model	Precision	Recall	F1	Method
TTPXHUNTER [19]	70.73	49.65	58.35	Classification-based
Finetuned-SecureBERT [15]	65.02	54.11	59.06	Classification-based
AttacKG [20]	9.75	76.36	17.29	Similarity-based
LADDER [25]	48.12	26.36	34.07	Similarity-based
ChatGPT-4o [30]	57.70	70.42	70.56	LLM-only
Our framework	**86.14**	**78.76**	**82.28**	LLM + Similarity

Since the dataset only contains labels for main techniques, all evaluations were performed accordingly, excluding sub-techniques. For both the single LLM baseline and our framework, we conducted five identical experiments and reported the average results. To reduce variance across runs, we set the temperature to 0 and the top_p parameter to 1 in each experiment. Table 5 presents the experimental results comparing our framework against five baseline models, each representing a different approach category. Our framework achieved the highest F1-score of 82.28%, consistently outperforming all baselines across precision and recall. Notably, even the LLM-only baseline (ChatGPT-4o) surpassed traditional encoder-based classifiers, indicating the strong potential of LLMs in cybersecurity applications. Across five runs, the standard deviation of the F1-score was approximately 0.3% for ChatGPT-4o and 0.2% for our framework, further demonstrating the stability of the reported results.

1) PERFORMANCE OF CLASSIFICATION-BASED METHODS

The classification-based baselines, TTPXHUNTER and Finetuned-SecureBERT, achieved relatively high precision (70.73% and 65.02%, respectively) but suffered from low recall (49.65% and 54.11%). This performance limitation resulted in lower scores than those reported in the original papers, which can be attributed to the variation in optimal thresholds across different evaluation datasets. As noted by Li et al. [16], using a fixed threshold without adaptation can lead to performance degradation, which may hinder the ability of such models to maintain stable performance in practical, real-world environments. Moreover, classification-based approaches require retraining whenever the MITRE ATT&CK framework is updated, posing challenges for long-term scalability and maintainability.

2) PERFORMANCE OF SIMILARITY-BASED METHODS

Similarity-based methods, AttacKG and LADDER, exhibited lower overall performance than other baselines due to their low precision. In particular, AttacKG recorded the lowest precision among all baselines at 9.75%, while maintaining a high recall of 76.36%, suggesting a tendency to over-extract TTPs and retrieve many irrelevant candidates. This method represents attack patterns as graphs, which often leads to structurally similar representations across different techniques. Such ambiguity in graph-level similarity results in excessive false positives. While LADDER, unlike AttacKG, performs similarity matching at the sentence level, it simplifies the input sentences during the attack pattern extraction process, which leads to a loss of information. Furthermore, a semantic mismatch exists between the extracted threat behaviors and the ontology entries it compares against. Specifically, while the extracted behaviors are close to MITRE ATT&CK procedures, LADDER aligns them with the technique descriptions, which are more abstract and high-level. This mismatch in semantic resolution degrades the performance of the model.

3) PERFORMANCE OF SINGLE LLM BASELINE

The LLM-only baseline, ChatGPT-4o, yielded a balanced performance with a recall of 70.42% and an F1-score of 70.56%, outperforming both encoder-based classifiers and similarity-based baselines. However, its precision remained relatively low (57.70%), which suggests that while LLMs have strong semantic reasoning capabilities, they may hallucinate and generate false positives. This characteristic limits its standalone applicability in practice, as false positives can significantly reduce overall system reliability.

4) PERFORMANCE OF OUR FRAMEWORK

To overcome the limitations observed in both classification-based and similarity-based methods, our framework adopts a hybrid approach that combines LLM-based threat behavior extraction with embedding-based similarity matching. To minimize the information loss that occurs during the attack pattern extraction process, as seen in similarity-based approaches (AttacKG, LADDER), our framework represents attack patterns in the form of original sentences and compares them directly with MITRE ATT&CK procedures to ensure semantic alignment. This alignment resolves the granularity mismatch seen in LADDER and contributes to high precision (86.14%). Furthermore, our framework addresses the hallucination issue in LLMs by constraining predictions within a similarity-based candidate TTPs. Specifically, the overall precision increases from 57.70% to 86.14%, reflecting a 28.44%p improvement. Additionally, our framework achieved the highest recall (78.76%) and overall F1-score (82.28%) among all models by splitting threat actions during extraction such that each action corresponds to a single technique.

A detailed analysis revealed that false positives most often occur when the *Validator* retains multiple semantically similar techniques generated by the *Technique Candidate Generator*. This happens when the system fails to narrow down to the single most appropriate one inferred from the purpose of the threat action. In contrast, false negatives occur when correct TTPs extracted by the *Technique Candidate Generator* are removed during validation. Considering the performance of our framework, it can be observed that recall is lower than precision, indicating that false negatives occur more frequently than false positives. The relatively higher precision compared to recall can be largely attributed to the *Validator*, and this effect is analyzed in detail in Section VI-C.

Unlike classification-based methods, it does not require retraining when a new version of MITRE ATT&CK is released. The system remains up-to-date by simply updating the STIX-based knowledge base, making it more scalable and maintainable in real-world cybersecurity environments.

VI. ABLATION STUDY

This section presents a series of ablation experiments to evaluate whether each component effectively fulfills its designated role within our proposed framework. These experiments analyze the performance of individual components and their contribution to the overall framework. The evaluation dataset and LLM used in this ablation study are identical to those described in Section V.

A. EXTRACTOR PERFORMANCE ANALYSIS

To analyze the impact of threat action extraction methods on our framework's performance, we compared three approaches:

- **Expert Extraction**: Threat actions manually extracted by cybersecurity professionals.
- **Direct Extraction Prompt**: The LLM is prompted to extract threat actions by identifying complete sentences in the document that describe attacker behavior, preserving their original structure without modification.
- **Atomic Reconstruction Prompt**: The LLM is instructed to extract threat actions while splitting sentences that describe multiple techniques into separate atomic statements, ensuring each extracted action corresponds to precisely one MITRE ATT&CK technique (the approach adopted in our *Extractor*).

For consistency, all extracted threat actions were processed by the same downstream modules which include the *Technique Candidate Generator* and *Validator*, so that the observed performance differences can be attributed solely to the extraction method. We conducted the evaluation on 46 documents from the benchmark dataset, and the results are summarized in Table 6. These findings underscore a clear correlation between the granularity of extracted threat actions and the overall effectiveness of our framework.

The direct extraction prompt yields high precision but suffers from reduced recall. Its superior precision (91.62%) compared to the atomic reconstruction prompt (86.14%) is attributed to differences in the number of extracted threat actions. Many sentences in CTI documents contain multiple threat actions corresponding to distinct MITRE ATT&CK techniques. For instance, a sentence describing both ''decrypting'' and ''decoding'' functionalities would be extracted as a single threat action using the direct extraction prompt, even though these actions map to separate techniques. In contrast, the atomic reconstruction prompt decomposes such sentences into multiple segments, each corresponding to a single TTP. This results in a greater number of extracted threat actions and, consequently, more queries sent to the *Technique Candidate Generator*. Although this expanded candidate space reduces precision, it significantly improves recall by enabling more comprehensive technique coverage. Since direct extraction applies no decomposition, it leads to incomplete coverage and lower recall. While atomic reconstruction trades some precision, the substantial gains in recall yield higher overall F1-scores, demonstrating its effectiveness in capturing a broader set of relevant techniques.

TABLE 6. Performance comparison of different threat action extraction methods.

Extraction Method	Precision	Recall	F1
Expert Extraction	80.37	74.31	77.22
Direct Extraction Prompt	**91.62**	59.93	72.46
Atomic Reconstruction Prompt	86.14	**78.76**	**82.28**

Expert extractions, while ensuring that identified sentences are definitively related to cyber threat actions, also exhibited performance limitations when applied within our framework. As threat actions are manually selected by analysts, the number of techniques covered by each extracted action is often inconsistent, depending on how broadly or narrowly the behavior is interpreted. This variability in the level of detail often results in lower recall and precision compared to our *Extractor*. In contrast, the atomic reconstruction prompt significantly improves recall (78.76%) by isolating distinct threat behaviors, each aligned with an individual technique. This technique-specific extraction enables the *Technique Candidate Generator* to create more precise semantic mappings, resulting in the highest F1-score (82.28%) among the three methods. The results demonstrate that our *Extractor*'s ability to decompose complex threat descriptions into atomic, technique-aligned statements is pivotal in maximizing the framework's overall performance.

B. ANALYSIS OF TECHNIQUE CANDIDATE GENERATOR

We analyzed the performance of the *Technique Candidate Generator* by evaluating three key aspects: (1) a validation of our sentence similarity-based approach against an LLM-based alternative, (2) a comparison of different sentence embedding models employed within the generator, and (3) the overall coverage of its retrieved candidate techniques.

TABLE 7. Performance comparison of candidate generation methods.

Candidate TTPs Generation Method	Recall
Extractor + LLM	66.09
Extractor + Technique Candidate Generator (Ours)	97.26

1) VALIDATION OF SENTENCE SIMILARITY-BASED APPROACH

To validate our proposed sentence similarity-based approach, we compared its performance against an alternative method that directly prompts an LLM to generate candidates based on threat actions extracted by the *Extractor*. The primary role of this generator component is to maximize recall, ensuring that all potentially relevant techniques are passed to the subsequent *Validator* component. Therefore, we focus on recall as the key metric in this comparison.

The results in Table 7 show that our proposed generator achieved a recall of 97.26%, which is significantly higher than the 66.09% achieved by the direct LLM-based approach. This indicates that our sentence embedding-based retrieval method is more effective at comprehensively identifying the full spectrum of relevant techniques from the knowledge base. By providing a more inclusive candidate set to the *Validator*, our approach minimizes the risk of prematurely discarding correct techniques. This outcome validates our two-step design: a high-recall generator that provides a robust foundation for the high-precision *Validator* component.

2) COMPREHENSIVE EVALUATION OF EMBEDDING METHODS

The selection of an appropriate embedding model is a critical design choice within the *Technique Candidate Generator*. To validate our use of a general-purpose LLM-based embedding (*text-embedding-ada-002*), we conducted a comprehensive performance comparison against a encoder-based baseline. This baseline was developed using the Sentence-Transformer (SBERT) framework [46], which is designed to derive semantically meaningful sentence embeddings through a domain-adaptive training strategy.

The fine-tuning process employed a triplet-based contrastive learning framework to optimize the embedding space for technique differentiation. Procedures from our knowledge base were grouped by their corresponding MITRE ATT&CK technique IDs into distinct sets, defined as follows:

$$\begin{aligned} &\mathcal{G} = \{t_i : S_i\}, \quad \text{where } t_i \text{ is the technique ID,} \\ &S_i = \{s_1, s_2, \ldots, s_n\} \text{ represents the subset} \\ &\qquad \times \text{ of the knowledge base associated with } t_i. \end{aligned} \tag{2}$$

To construct meaningful triplets (s_a, s_p, s_n) for contrastive learning, we define each component as follows:

- **Anchor sentence** (s_a): A randomly selected sentence from a specific technique category S_i.
- **Positive sentence** (s_p): Another sentence from the same category S_i, ensuring semantic similarity.

TABLE 8. Comparison of Technique Candidate Generator performance using LLM-based embeddings versus encoder-based embeddings.

Embedding Methods	Precision	Recall	F1
Finetuned Sentence-Transformer	37.29	98.97	54.17
OpenAI Embedding (Ours)	68.77	97.26	80.57

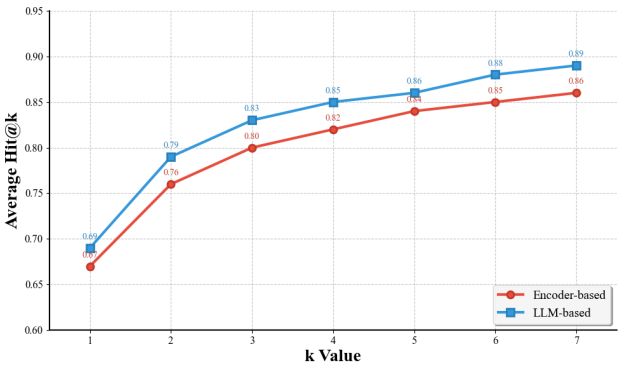


FIGURE 8. Retrieval effectiveness, measured by Average Hit@k, of the two embedding methods. The LLM-based embedding consistently outperforms the encoder-based model across all tested k values.

- **Negative sentence** (s_n): A sentence sampled from a different technique category S_j where $i \neq j$, ensuring semantic dissimilarity.

Thus, the triplet training dataset, $\mathcal{T}$, is structured as:

$$\mathcal{T} = \{(s_a, s_p, s_n) \mid s_a, s_p \in S_i, s_n \in S_j, i \neq j\}. \tag{3}$$

For this purpose, we utilized the *paraphrase-multilingual-MiniLM-L12-v2* model [47] and fine-tuned it using a Triplet Loss function, defined as:

$$\mathcal{L} = \max(0, d(s_a, s_p) - d(s_a, s_n) + \alpha), \tag{4}$$

where $d(x, y)$ represents the Euclidean distance between sentence embeddings, and α is a margin hyperparameter set to 5. Fine-tuning was performed with PyTorch using a batch size of 16 and the AdamW optimizer.

After establishing the encoder-based baseline, we conducted two experiments to compare its effectiveness with that of the LLM-based embedding. The first experiment evaluated the quality of the candidate sets generated by each model before the *Validator* stage. As shown in Table 8, both methods achieved high recall (97.26% and 98.97%, respectively), meeting the generator's primary goal. However, the LLM-based embedding showed a significantly better balance between precision and recall, improving precision by 31.48%p at the cost of only a 1.71%p decrease in recall. The resulting F1-score further confirms this advantage, with the LLM-based approach achieving 80.57% compared to 54.17% for the encoder-based method.

The second experiment evaluated the effectiveness of each embedding model in retrieving relevant procedures from the MITRE ATT&CK knowledge base, using the average hit@k metric to measure similarity search performance.

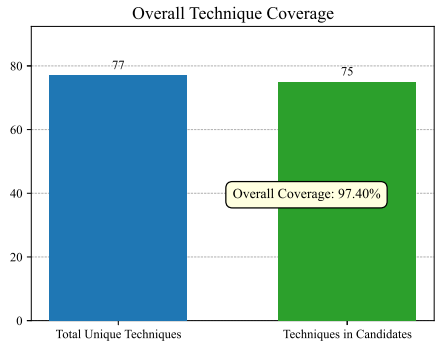


FIGURE 9. Overall technique coverage achieved by our *Technique Candidate Generator*. The chart shows the total number of unique techniques in the benchmark dataset and how many of these techniques are successfully included in candidate sets.

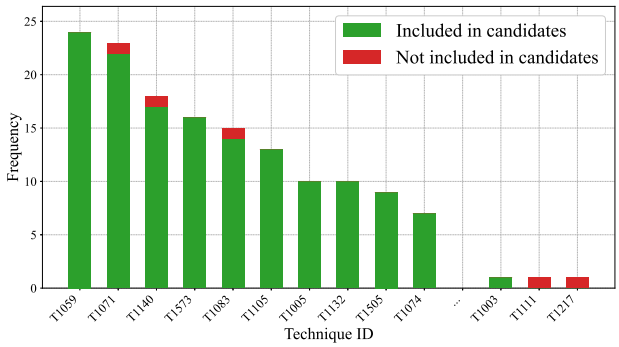


FIGURE 10. Coverage status of techniques sorted by frequency. Green bars represent techniques included in candidate sets, while red bars indicate techniques not captured by the *Technique Candidate Generator*. The height of each bar shows the frequency of the technique in the benchmark dataset.

As illustrated in Fig. 8, the LLM-based embedding consistently outperformed the fine-tuned encoder-based model across all values of k. Notably, the general-purpose OpenAI embeddings, despite not being specifically trained for this task, captured technique-specific semantics more effectively within threat procedure descriptions.

3) COVERAGE ANALYSIS OF CANDIDATE TECHNIQUES

We assess the coverage of techniques retrieved by the *Technique Candidate Generator* to determine how comprehensively the embedding-based retrieval method identifies techniques defined in the MITRE ATT&CK framework.

The overall technique coverage achieved by our framework is illustrated in Fig. 9. Out of 77 unique techniques present in the evaluation corpus, 75 techniques (97.4%) were successfully included in the candidate sets generated by our embedding-based retrieval method. This high coverage rate demonstrates the effectiveness of our approach in capturing the diverse range of techniques present in the benchmark dataset. Fig. 10 provides a more detailed view of the coverage for techniques sorted by their frequency in the dataset. The bars illustrate that our approach successfully captures the majority of commonly occurring techniques. Most high-frequency techniques are predominantly green, indicating their successful inclusion in our candidate sets. Only minimal red portions are observed for techniques like "T1071" and "T1140", which represent occasional missed instances. The only techniques entirely missing from our candidate sets were "T1111 (Multi-Factor Authentication Interception)" and "T1217 (Browser Information Discovery)", each of which appeared only once in the entire dataset. This limitation can be attributed to the insufficient representation of these techniques in our knowledge base, as they had few associated procedures for the embedding model to learn from. This finding suggests that the performance of our approach is directly related to the richness of procedures available in the knowledge base for each technique.

The high coverage achieved by our *Technique Candidate Generator* is particularly important for the success of our framework. By ensuring that almost all relevant techniques are available in the candidate techniques, this component provides the *Validator* with a comprehensive set of options from which to make its final selections, contributing significantly to the overall accuracy of our technique extraction system.

TABLE 9. Performance comparison of technique extraction across different *Validator* configurations.

Pipeline Configuration	Precision	Recall	F1
w/o *Validator*	68.77	97.26	80.57
w/ *Validator* (non-ranking)	79.60	68.15	73.43
w/ *Validator* (ranking)	86.14	78.76	82.28

C. ANALYSIS OF VALIDATOR

An ablation study was conducted to compare system performance with and without the *Validator* component, as well as with ranking-based versus non-ranking prompts, where the non-ranking prompt simply asks the LLM to filter the correct TTPs without performing a likelihood-based ranking process. This experiment aims to measure the impact of the *Validator* in selecting the correct technique IDs from the candidate set produced by the *Technique Candidate Generator*.

As demonstrated in Table 9, omitting the *Validator* yields a recall of 97.26% but limits precision to 68.77%, resulting in numerous false positives. Enabling the *Validator* with a non-ranking prompt increases precision to 79.60% but recall decreases to 68.15%, yielding an F1-score of 73.43%. When using the ranking-based prompt, precision further rises to 86.14% while recall reaches 78.76%, with an overall F1-score improving from 80.57% (w/o *Validator*) to 82.28%.

By instructing the LLM to rank candidate techniques based on their likelihood and contextual relevance, the *Validator* performs the selective filtering required for automated technique extraction. This precision-oriented shift demonstrates that the *Validator* effectively fulfills its intended purpose by emulating the human verification process, which prioritizes precision by filtering out incorrect candidates while limiting

the loss of relevant techniques. This improvement in precision enhances the reliability of the extracted TTPs, thereby making our framework more applicable in real-world security operations.

VII. CONCLUSION

In this paper, we proposed a framework that leverages LLMs to extract MITRE ATT&CK techniques from unstructured CTI documents with high F1-score and balanced precision-recall performance. The framework consists of three components: an *Extractor* for identifying threat actions, a *Technique Candidate Generator* for retrieving semantically similar techniques, and a *Validator* for final technique selection. An important observation from our study is that even a single LLM, using basic prompting, can achieve performance comparable to existing TTP extraction methods. Building on this baseline, our framework utilizes LLMs iteratively and incorporates embedding-based retrieval to mitigate hallucination by constraining LLM predictions within retrieved techniques. This multi-step approach led to notable improvements in F1-score and overall extraction performance, demonstrating its effectiveness for real-world CTI analysis. In our future work, we will explore computational efficiency by developing smaller, specialized LLMs focused specifically on attack behavior extraction and validation tasks, making our approach more cost-effective for practical deployments.

APPENDIX

PER-COMPONENT RUNTIME ANALYSIS

We evaluated the runtime of each component in our framework using the CTI-bench dataset, which contains 46 samples. Each experiment was repeated five times, yielding a total of 230 runs. For the LLM-based modules, the ChatGPT-4o model was accessed through the OpenAI API, and the embedding step was performed using *text-embedding-ada-002*. The *Extractor* required an average of 1.79 seconds, processing 426.5 tokens per run. The *Technique Candidate Generator* required an average of 3.12 seconds, with an average input size of 110 tokens. The *Validator* required an average of 2.78 seconds, processing 520 tokens per run.

REFERENCES

[1] CSO Online. *Stuxnet Explained: The First Known Cyberweapon*. Accessed: Feb. 22, 2025. [Online]. Available: https://www.csoonline.com/article/562691/stuxnet-explained-the-first-known-cyberweapon.html

[2] NPR. *A 'Worst Nightmare' Cyberattack: The Untold Story of the SolarWinds Hack*. Accessed: Feb. 22, 2025. [Online]. Available: https://www.npr.org/2021/04/16/985439655/a-worst-nightmare-cyberattack-the-untold-story-of-the-solarwinds-hack

[3] K. Scarfone and P. Mell, "Guide to intrusion detection and prevention systems (IDPS)," Nat. Inst. Standards Technol., Gaithersburg, MD, USA, Tech. 800-94, 2007. Accessed: Feb. 22, 2025. [Online]. Available: https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-94.pdf

[4] S. Krishnapriya and S. Singh, "A comprehensive survey on advanced persistent threat (APT) detection techniques," *Comput., Mater. Continua*, vol. 80, no. 2, pp. 2675–2719, 2024.

[5] M. R. Rahman, R. M. Hezaveh, and L. Williams, "What are the attackers doing now? Automating cyberthreat intelligence extraction from text on pace with the changing threat landscape: A survey," *ACM Comput. Surveys*, vol. 55, no. 12, pp. 1–36, Dec. 2023.

[6] The MITRE Corporation. *MITRE ATT&CK*. The MITRE Corporation. Accessed: Feb. 23, 2025. [Online]. Available: https://attack.mitre.org

[7] E. M. Hutchins, M. J. Cloppert, and R. M. Amin, "Intelligence-driven computer network defense informed by analysis of adversary campaigns and intrusion kill chains," Lockheed Martin, Bethesda, MD, USA, White Paper no., 2011. Accessed: Feb. 23, 2025. [Online]. Available: https://www.lockheedmartin.com/content/dam/lockheed-Martin/rms/documents/cyber/LM-White-Paper-Intel-Driven-Defense.pdf

[8] M. Tatam, B. Shanmugam, S. Azam, and K. Kannoorpatti, "A review of threat modelling approaches for APT-style attacks," *Heliyon*, vol. 7, no. 1, Jan. 2021, Art. no. e05969, doi: 10.1016/j.heliyon.2021.e05969.

[9] F. Menges and G. Pernul, "A comparative analysis of incident reporting formats," *Comput. Secur.*, vol. 73, pp. 87–101, Mar. 2018.

[10] OASIS Open. *Structured Threat Information Expression (STIX)*. Accessed: Feb. 23, 2025. [Online]. Available: https://oasis-open.github.io/cti-documentation/stix/intro

[11] Z. Zhu and T. Dumitraş, "FeatureSmith: Automatically engineering features for malware detection by mining the security literature," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Oct. 2016, pp. 767–778.

[12] V. Legoy, M. Caselli, C. Seifert, and A. Peter, "Automated retrieval of ATT&CK tactics and techniques for cyber threat reports," 2020, *arXiv:2004.14322*.

[13] M. R. Rahman and L. Williams, "From threat reports to continuous threat intelligence: A comparison of attack technique extraction methods from textual artifacts," 2022, *arXiv:2210.02601*.

[14] L. Zongxun, L. Yujun, Z. Haojie, and L. Juan, "Construction of TTPS from APT reports using bert," in *Proc. 18th Int. Comput. Conf. Wavelet Act. Media Technol. Inf. Process. (ICCWAMTIP)*, Dec. 2021, pp. 260–263.

[15] V. Orbinato, M. Barbaraci, R. Natella, and D. Cotroneo, "Automatic mapping of unstructured cyber threat intelligence: An experimental study: (Practical experience report)," in *Proc. IEEE 33rd Int. Symp. Softw. Rel. Eng. (ISSRE)*, Oct. 2022, pp. 181–192, doi: 10.1109/issre55969.2022.00027.

[16] L. Li, C. Huang, and J. Chen, "Automated discovery and mapping ATT&CK tactics and techniques for unstructured cyber threat intelligence," *Comput. Secur.*, vol. 140, May 2024, Art. no. 103815, doi: 10.1016/j.cose.2024.103815.

[17] The MITRE Corporation. *MITRE Caldera: Automated Adversary Emulation Platform*. The MITRE Corporation. Accessed: Feb. 25, 2025. [Online]. Available: https://github.com/mitre/caldera

[18] Center for Threat-Informed Defense. *Threat Report ATT&CK Mapper (TRAM)*. MITRE Engenuity. Accessed: Feb. 26, 2025. [Online]. Available: https://github.com/mitre-attack/tram

[19] N. Rani, B. Saha, V. Maurya, and S. K. Shukla, "TTPXHunter: Actionable threat intelligence extraction as TTPs from finished cyber threat reports," *Digit. Threats, Res. Pract.*, vol. 5, no. 4, pp. 1–19, Dec. 2024.

[20] Z. Li, J. Zeng, Y. Chen, and Z. Liang, "AttacKG: Constructing technique knowledge graph from cyber threat intelligence reports," in *Proc. Eur. Symp. Res. Comput. Secur.*, 2022, pp. 589–609, doi: 10.1007/978-3-031-17140-6_29.

[21] M. T. Alam, D. Bhusal, L. Nguyen, and N. Rastogi, "CTIBench: A benchmark for evaluating LLMs in cyber threat intelligence," 2024, *arXiv:2406.07599*.

[22] G. Husari, E. Al-Shaer, M. Ahmed, B. Chu, and X. Niu, "TTPDrill: Automatic and accurate extraction of threat actions from unstructured text of CTI sources," in *Proc. 33rd Annu. Comput. Secur. Appl. Conf.*, Dec. 2017, pp. 103–115.

[23] E. Aghaei, X. Niu, W. Shadid, and E. Al-Shaer, "SecureBERT: A domain-specific language model for cybersecurity," in *Proc. Int. Conf. Secur. Privacy Commun. Syst.*, 2022, pp. 39–56.

[24] I. Beltagy, K. Lo, and A. Cohan, "SciBERT: A pretrained language model for scientific text," 2019, *arXiv:1903.10676*.

[25] M. T. Alam, D. Bhusal, Y. Park, and N. Rastogi, "Looking beyond IoCs: Automatically extracting attack patterns from external CTI," in *Proc. 26th Int. Symp. Res. Attacks, Intrusions Defenses*, Oct. 2023, pp. 92–108.

[26] R. Fayyazi, R. Taghdimi, and S. J. Yang, "Advancing TTP analysis: Harnessing the power of large language models with retrieval augmented generation," in *Proc. Annu. Comput. Secur. Appl. Conf. Workshops (ACSAC Workshops)*, Dec. 2024, pp. 255–261.

[27] MITRE Corporation. (2024). *CTI: Cyber Threat Intelligence Repository*. MITRE Corporation. Accessed: Mar. 2, 2025. [Online]. Available: https://github.com/mitre/cti/tree/master
[28] MISP Project. *MISP–Malware Information Sharing Platform & Threat Sharing*. Accessed: Feb. 22, 2025. [Online]. Available: https://www.misp-project.org
[29] Filigran. *OpenCTI–Open Cyber Threat Intelligence Platform*. Accessed: Feb. 26, 2025. [Online]. Available: https://www.opencti.io/
[30] OpenAI. *GPT-4o Model Documentation*. Accessed: Apr. 12, 2025. [Online]. Available: https://platform.openai.com/docs/models/gpt-4o
[31] G. Tsoumakas, I. Katakis, and I. Vlahavas, "Mining multi-label data," in *Data Mining and Knowledge Discovery Handbook*, 2nd ed. New York, NY, USA: Springer, 2010, pp. 667–685.
[32] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, "Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing," *ACM Comput. Surveys*, vol. 55, no. 9, pp. 1–35, Sep. 2023.
[33] H. Ji, J. Yang, L. Chai, C. Wei, L. Yang, Y. Duan, Y. Wang, T. Sun, H. Guo, T. Li, C. Ren, and Z. Li, "SEvenLLM: Benchmarking, eliciting, and enhancing abilities of large language models in cyber threat intelligence," 2024, *arXiv:2405.03446*.
[34] P. Ranade, A. Piplai, A. Joshi, and T. Finin, "CyBERT: Contextualized embeddings for the cybersecurity domain," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2021, pp. 3334–3342.
[35] T. Brown, B. Mann, and N. Ryder, "Language models are few-shot learners," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 1877–1901.
[36] W. X. Zhao et al., "A survey of large language models," 2023, *arXiv:2303.18223*.
[37] J. Zhang, H. Bu, H. Wen, Y. Liu, H. Fei, R. Xi, L. Li, Y. Yang, H. Zhu, and D. Meng, "When LLMs meet cybersecurity: A systematic literature review," *Cybersecurity*, vol. 8, no. 1, pp. 1–41, Feb. 2025, doi: 10.1186/s42400-025-00361-w.
[38] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Human Lang. Technol.*, 2019, pp. 4171–4186.
[39] Google DeepMind. *Gemini Models*. Accessed: Dec. 30, 2024. [Online]. Available: https://deepmind.google/technologies/gemini/pro/
[40] Meta. *Meta-Llama-3-8B*. Accessed: Dec. 30, 2024. [Online]. Available: https://huggingface.co/meta-llama/Meta-Llama-3-8B
[41] Meta. (2024). *Meta-Llama-3-70B*. Accessed: Feb. 22, 2025. [Online]. Available: https://huggingface.co/meta-llama/Meta-Llama-3-70B
[42] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.
[43] OASIS Cyber Threat Intelligence Technical Committee. *CTI-Python-STIX2*. Accessed: Feb. 22, 2025. [Online]. Available: https://github.com/oasis-open/cti-python-stix2
[44] OpenAI. *Text-Embedding-ADA-002*. Accessed: Feb. 22, 2025. [Online]. Available: https://platform.openai.com/docs/guides/embeddings
[45] M. Douze, A. Guzhva, C. Deng, J. Johnson, G. Szilvasy, P.-E. Mazaré, M. Lomeli, L. Hosseini, and H. Jégou, "The faiss library," 2024, *arXiv:2401.08281*.
[46] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," 2019, *arXiv:1908.10084*.
[47] N. Reimers. (2021). *Paraphrase-Multilingual-MiniLM-L12-V2*. Accessed: Feb. 22, 2025. [Online]. Available: https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2



HYOUNG ROK KIM received the B.E. degree in cyber defense and the M.S. degree in cybersecurity from Korea University, Seoul, Republic of Korea, in 2019 and 2025, respectively. He was with the Agency for Defense Development, where he contributed to projects on AI-based cyber attack defense methods and attack simulation. Currently, he is an Army Captain with the ROK Cyber Operations Command (ROKCC), engaging in cybersecurity tasks centered on penetration testing, security assessments, and AI. His research interests include threat hunting and AI-based cybersecurity and red teaming.



DONGHYEON LEE received the B.S. degree in physics from Korea University, Seoul, South Korea, in 2024, where he is currently pursuing the M.S. degree with the Department of Computer Science. He is a Researcher with AIGEN Sciences, where he is involved in projects related to drug discovery, pharmaceutical patent processing, and the development of LLM-based medical chatbots. His research interests include biomedical large language models and reasoning compression.



INSUP LEE received the B.E. degree in cyber defense from Korea University, Seoul, Republic of Korea, in 2018, where he is currently pursuing the Ph.D. degree in cybersecurity. From 2018 to 2023, he was a Researcher with the Cyber Technology Center, Agency for Defense Development, Republic of Korea. His research interests include generative models, AI-driven security, adversarial machine learning, and secure communications.



SOOHAN LEE received the B.E. degree in cyber defense and the M.S. degree in cybersecurity from Korea University, in 2020 and 2023, respectively. He was a Red Team Operator and a Cyber Training Instructor with Republic of Korea Cyber Command (ROKCC). He was also a Researcher with the Cyber Technology Center, Agency for Defense Development (ADD). Currently, he has rejoined ROKCC as a Researcher, focusing on cybersecurity and artificial intelligence. His research interests include artificial intelligence, usable security, security engineering, and red teaming.



SANGJIN LEE received the Ph.D. degree from the Department of Mathematics, Korea University, in 1994. From 1989 to 1999, he was a Senior Researcher with the Electronics and Telecommunications Research Institute, South Korea. He has been running the Digital Forensic Research Center, Korea University, since 2008. He is currently the President of the Division of Information Security, Korea University. He has authored or co-authored more than 130 papers in various archival journals and conference proceedings and more than 200 articles in domestic journals. His research interests include digital forensics, data processing, forensic framework, and incident response.